WORKED SOLUTION

STEP II 1993, Question 5

Pure • Polar Curves and Arc Length

STEP II 1993, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q5

The key idea

The right angle at R lets us read off every length in the small triangle PQR from the right-angled triangle ORQ, using only OQ and the small angle $\delta\theta$. Taking the limit as $Q \to P$ turns the chord PQ into the tangent. For the spiral, $\mathrm{d}r/\mathrm{d}\theta$ is a constant multiple of r, so $r\,\mathrm{d}\theta/\mathrm{d}r$ is the same everywhere, and the arc-length integrand is just an exponential.

Solution

The angle between the curve and the radius

The point R lies on the line OP (produced), and $\angle ORQ$ is a right angle. In the right-angled triangle ORQ the hypotenuse is $OQ = r + \delta r$ and the angle at O is $\delta\theta$, so

$$QR = (r+\delta r)\sin\delta\theta, \qquad OR = (r+\delta r)\cos\delta\theta.$$

Since $OP = r$, we have $PR = OR - OP = (r+\delta r)\cos\delta\theta - r$. The triangle PRQ is right-angled at R, so

$$\tan\angle QPR = \frac{QR}{PR} = \frac{(r+\delta r)\sin\delta\theta}{(r+\delta r)\cos\delta\theta - r}.$$

For small $\delta\theta$ we have $\sin\delta\theta = \delta\theta + O(\delta\theta^3)$ and $\cos\delta\theta = 1 + O(\delta\theta^2)$, so

$$\tan\angle QPR = \frac{r\,\delta\theta + \delta r\,\delta\theta + \cdots}{\delta r + O(\delta\theta^2)} = \frac{r\dfrac{\delta\theta}{\delta r} + \delta\theta + \cdots}{1 + O\left(\dfrac{\delta\theta^2}{\delta r}\right)},$$

where the dots are terms that are small compared with those shown. As $Q \to P$, both δr and $\delta\theta$ tend to 0 with $\delta\theta/\delta r \to \mathrm{d}\theta/\mathrm{d}r$. So $\delta\theta^2/\delta r = \delta\theta\cdot(\delta\theta/\delta r) \to 0$, and also $\delta\theta \to 0$. Hence

$$\tan\angle QPR \to r\frac{\mathrm{d}\theta}{\mathrm{d}r}.$$

Ask yourself Why is the limit of $\angle QPR$ the angle at which the curve cuts OP?

The angle QPR is the angle between the chord PQ and the radius line through P. As $Q \to P$ along the curve, the chord PQ turns into the tangent at P. So in the limit $\angle QPR$ becomes the angle between the tangent and OP, which is what "the angle at which the curve cuts OP" means. Therefore that angle is

$$\tan^{-1}\left(r\frac{\mathrm{d}\theta}{\mathrm{d}r}\right),$$

as required.

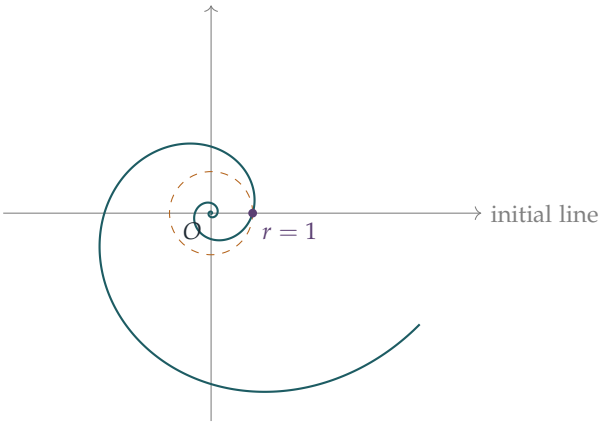
The equiangular spiral

For $r = \mathrm{e}^{\theta\cot\alpha}$,

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = \cot\alpha\,\mathrm{e}^{\theta\cot\alpha} = r\cot\alpha, \qquad \text{so} \qquad r\frac{\mathrm{d}\theta}{\mathrm{d}r} = \frac{r}{r\cot\alpha} = \tan\alpha.$$

Because $0 < \alpha < \frac{1}{2}\pi$, the angle α lies in the range of $\tan^{-1}$, so $\tan^{-1}(\tan\alpha) = \alpha$. This doesn't depend on θ, so the spiral cuts every radius line at the angle α, at every point.

Sketch. Since $\cot\alpha > 0$, r increases steadily with θ. Each full turn multiplies r by the fixed factor $\mathrm{e}^{2\pi\cot\alpha}$. As $\theta \to -\infty$, $r \to 0$, so the curve winds round the origin infinitely often without ever reaching it. As $\theta \to +\infty$, it spirals outwards anticlockwise. It crosses the initial line at $r = 1$ when $\theta = 0$. The sketch uses $\cot\alpha = 0.3$.



The dashed circle is $r = 1$. The part of the spiral inside it is the infinite inward coil.

Arc lengths

Using the given formula and $\mathrm{d}r/\mathrm{d}\theta = r\cot\alpha$,

$$\left(\frac{\mathrm{d}s}{\mathrm{d}\theta}\right)^2 = r^2 + r^2\cot^2\alpha = r^2\operatorname{cosec}^2\alpha, \qquad \text{so} \qquad \frac{\mathrm{d}s}{\mathrm{d}\theta} = \frac{\mathrm{e}^{\theta\cot\alpha}}{\sin\alpha},$$

taking the positive root, since s increases with θ and $\sin\alpha > 0$.

The complete turn starting at $r = 1$ and going outwards runs from $\theta = 0$ to $\theta = 2\pi$. Its length is

$$\int_0^{2\pi} \frac{\mathrm{e}^{\theta\cot\alpha}}{\sin\alpha}\,\mathrm{d}\theta = \frac{1}{\sin\alpha}\cdot\frac{\mathrm{e}^{2\pi\cot\alpha}-1}{\cot\alpha} = \sec\alpha\left(\mathrm{e}^{2\pi\cot\alpha}-1\right).$$

The part with $r \le 1$ is $\theta \le 0$, that is, $-\infty < \theta \le 0$. Its length is the improper integral

$$\int_{-\infty}^{0} \frac{\mathrm{e}^{\theta\cot\alpha}}{\sin\alpha}\,\mathrm{d}\theta = \lim_{T\to\infty}\frac{1-\mathrm{e}^{-T\cot\alpha}}{\sin\alpha\cot\alpha} = \sec\alpha,$$

since $\cot\alpha > 0$ makes $\mathrm{e}^{-T\cot\alpha} \to 0$. So even though this part winds round the origin infinitely many times, its total length is finite, namely $\sec\alpha$.

Checks

- As $\alpha \to \frac{1}{2}\pi$ we have $\cot\alpha \to 0$, and the spiral becomes the unit circle, which every radius cuts at a right angle. The length of one turn is $\sec\alpha(\mathrm{e}^{2\pi\cot\alpha}-1) \approx \sec\alpha\cdot 2\pi\cot\alpha = 2\pi\operatorname{cosec}\alpha \to 2\pi$, the circumference.
- A nice geometric check of $\sec\alpha$: the length from any point to the origin along the spiral is $r\sec\alpha$, which is the length of the tangent segment from that point to the line through O perpendicular to OP.
- Numerically, with $\alpha = 0.9$, summing 2×10^5 chords gives one turn of length 233.8349 and the inner part of length 1.608726. These agree with $\sec\alpha(\mathrm{e}^{2\pi\cot\alpha}-1)$ and $\sec\alpha$ to at least eight significant figures.

Common slips

- Writing $PR = \delta r$ and $QR = r\,\delta\theta$ without explaining why. Both are only true to first order, and the limit argument is what the question is testing.
- Integrating over $[0, 2\pi]$ for the inner part instead of over $(-\infty, 0]$. The part with $r \le 1$ consists of infinitely many turns, not one.

Beyond STEP

The equiangular (logarithmic) spiral is self-similar: scaling it by $\mathrm{e}^{2\pi\cot\alpha}$, or by any factor, gives the same curve rotated. In complex terms it is $z = \mathrm{e}^{(\cot\alpha+\mathrm{i})\theta}$, the image of a straight line under the exponential map, and the constant angle follows from the fact that exp is conformal (it preserves angles). Needham's *Visual Complex Analysis* develops this picture beautifully. The finite length of an infinitely coiled curve is a first taste of the improper integrals and convergence questions met in a first analysis course.

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